

M2-Thesis

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Preface

Throughout my undergraduate and graduate studies in mathematics I have been caught between two main interests: the applied world of machine learning, and the theoretical world of topology and geometry. While this is often viewed as an unusual mix to choose, I have always found it greatly enjoyable.

This culminated in me wanting to answer the question of what is happening to the “shape” of data throughout a machine learning process, and also what even is the “shape” of data.

The value in understanding this is twofold: first, to be able to make better models by leveraging our understanding of how they work, and second, to be able to understand how to interpret what these models are doing. The dance between naturally arising data and the model trying to fit without knowing how the data is generated is a curious one. My main motivation was to try to understand this.

One big idea is the notion of data lying on some low-dimensional space, which makes both learning and sampling from distributions supported on this lower-dimensional space easier. However this belief only truly simplifies the system in the small noise regime. All sorts of complex geometry can happen in the higher noise regimes. We do not, as we will see, have a nicely behaved interpolating family of manifolds.

Before doing the IASD M2 program, I had written a master’s dissertation on diffusion models under the manifold hypothesis, under the supervision of George Deligiannidis. During this I became deeply interested in dimension estimation of datasets. This at best vague and at worse terribly ill-posed question seemed to me one where an answer would tell us a lot about the natural geometry of data, and how diffusion models are able to make use of this. Almost every paper I found on the interplay between manifolds and diffusion models cited work from Eddie Aamari.

In November 2025 I reached out to Eddie Aamari to further discuss what the geometry of data could look like, and this led to this internship project. The initial goal was to understand the interplay between geometry affecting sampling error and how this could encourage simpler embeddings in a latent diffusion setting. However, given the complexity of work being done limited time was spent on investigating autoencoders so far.

A key challenge of this project has been striking the balance between geometry: where things are well defined, and diffusion of probability densities: where simply by adding noise we violently change the notions of geometry we would like to equip our density with. This tradeoff has led to looking at several objects, which while interesting in their own right have not been helpful in bounding sampling error. We do think the final part on posterior covariance fields [section 6](#) does provide new insight on where true complexity lies in the diffusion process.

Abstract

This is the abstract of the thesis.

1 Introduction

We started by looking at ([Potapchik et al., 2024](#)).

2 Background

3 Anisotropic Stability Bound

4 Ridges

5 Level Sets

6 Posterior Covariance Field

Here we introduce the ultimate object we analyse: posterior covariance fields.

Definition 6.1. Let $Y_t = X + \sqrt{t}Z$, $m_t(y) := \mathbb{E}[X \mid Y_t = y]$, and

$$A_t(y) := \frac{1}{t} \text{Cov}(X \mid Y_t = y).$$

We call the weighted object $\mathfrak{F}_t := \text{Law}(Y_t, m_t(Y_t), A_t(Y_t))$ the *posterior covariance field*.

Note: while we are also including the posterior mean $m_t(y)$ in our definition, we keep the name for simplicity. The posterior mean m_t and by extension the score s_t is the object whose regularity we want to understand, so its inclusion is natural.

There are a few reasons why it makes sense to look at this object. First, it is not plagued by the issues we encountered in the analysis of ridges or level sets in [section 4](#) and [section 5](#). These objects suffered from being too geometric for our use case, where we care about noise scales much greater than 0. We need a more analytic quantity, and the posterior covariance field is defined for all noise scales $t > 0$. Second, the posterior covariance captures the local geometry of the score. The posterior covariance corresponds to the Jacobian of the score. This link is given in [section 2](#). Here we normalise the posterior covariance by t to account for the scale of noise added. Third, here we deal with expectations, rather than L^∞ -type quantities like the reach τ_M . If there exists a region of the density which is geometrically complex and as a consequence has poor regularity, but which carries almost no mass, we should not let it dramatically change our understanding of the behaviour of the score.

6.1 Three functionals of the spectrum

All three quantities below are moments of the same object: the spectral measure of A_t , averaged over $y \sim p_t$. Write $a_1(y), \dots, a_D(y)$ for the eigenvalues of $A_t(y)$ and let $\mu_t := \text{Law}(a_I(Y_t))$ with I uniform on $\{1, \dots, D\}$, so that $\mathbb{E} \sum_i f(a_i) = D \int f d\mu_t$ for any f .

6.1.1 Effective dimension

Definition 6.2.

$$d_{\text{eff}}(t) := \mathbb{E}[\text{tr } A_t(Y_t)] = \mathbb{E} \sum_i a_i. \quad (1)$$

Proposition 6.3 (Exact MMSE identity). *Let $\text{mmse}(t) := \mathbb{E}|X - m_t(Y_t)|^2$. Then*

$$d_{\text{eff}}(t) = \frac{\text{mmse}(t)}{t}. \quad (2)$$

Proof. $\text{tr Cov}(X \mid Y_t) = \mathbb{E}[|X - m_t(Y_t)|^2 \mid Y_t]$; take expectations and divide by t . $\square$

Identity (2) is exact, not asymptotic, and has two consequences of practical importance. First, d_{eff} is dimensionless: it is residual posterior uncertainty measured in units of injected noise variance. Second, $\text{mmse}(t)$ is precisely the population value of the ε -prediction (or x_0 -prediction) denoising loss at noise level t , so d_{eff} requires no computation beyond the training objective already being evaluated.

Link to dimension. An orthogonal projector of rank k has spectrum $\{1^{(k)}, 0^{(D-k)}\}$ and trace k . In the ideal manifold limit $A_t \approx P_{T_x \mathcal{M}}$ and hence $d_{\text{eff}}(t) \approx d$. Equivalently, via (2): a d -dimensional manifold leaves $O(t)$ posterior uncertainty in each of d tangent directions and $o(t)$ in the normal directions, so $\text{mmse} \approx d \times t$. Each eigenvalue may be read as a fractional dimension: $a_i = 0.4$ contributes 0.4 effective directions.

Failure mode. d_{eff} is a first moment and therefore cannot distinguish $\text{diag}(1, 1, 1, 0, 0, 0)$ from $\text{diag}(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$: both have trace 3. It also over-counts when $a_i > 1$, in which case the “fractional dimension” reading breaks down entirely. The next two functionals repair exactly these two deficiencies.

6.1.2 Distance to projector structure

Definition 6.4.

$$\Delta_{\text{proj}}(t) := \mathbb{E} \sum_i \min\{a_i^2, (a_i - 1)^2\}. \quad (3)$$

The summand is the squared distance from a_i to the set $\{0, 1\}$, and the two branches exchange at $a_i = \frac{1}{2}$:

$$\min\{a^2, (a - 1)^2\} = \begin{cases} a^2, & a \leq \frac{1}{2} \quad (\text{read as normal-like}), \\ (a - 1)^2, & a \geq \frac{1}{2} \quad (\text{read as tangent-like}). \end{cases} \quad (4)$$

Proposition 6.5. Let $\mathcal{P} = \{P \in \mathbb{R}^{D \times D} : P = P^\top = P^2\}$ be the set of orthogonal projectors and let A be symmetric with eigenvalues a_i . Then

$$\min_{P \in \mathcal{P}} \|A - P\|_F^2 = \sum_i \min\{a_i^2, (a_i - 1)^2\}, \quad \text{hence} \quad \Delta_{\text{proj}}(t) = \mathbb{E}[\text{dist}_F^2(A_t(Y_t), \mathcal{P})]. \quad (5)$$

Proof. $\mathcal{P}$ is invariant under conjugation by orthogonal matrices, and for symmetric A the Frobenius distance to a conjugation-invariant set is minimized by a P sharing the eigenvectors of A (a consequence of the Hoffman–Wielandt inequality). It then remains to round each a_i to the nearer of 0 or 1. $\square$

Thus Δ_{proj} answers: *how far, on average, is the soft posterior geometry from being a genuine tangent-space projector?* It vanishes exactly when $A_t(Y_t)$ is a projector almost surely.

Remark 6.6 (Interpretation as a resolution probe). For data concentrated near a manifold but with normal thickness δ , the normal eigenvalues behave as $a_n \approx \delta^2/(\delta^2 + t)$, which passes through $\frac{1}{2}$ at $t \approx \delta^2$ and contributes $\frac{1}{4}$ to Δ_{proj} there. Consequently $\Delta_{\text{proj}}(t)$ peaks at the scale at which the manifold hypothesis stops being exactly true, and its peak location is a readout of that scale. Δ_{proj} is best regarded as a scale-resolved test of manifold-likeness rather than a single verdict.

6.1.3 Ambiguity

Definition 6.7. With $(x)_+ := \max\{x, 0\}$,

$$\mathcal{A}(t) := \mathbb{E} \sum_i (a_i - 1)_+^2. \quad (6)$$

The motivation is dynamical rather than geometric. $a_i > 1$ occurs if and only if the reverse probability-flow ODE is locally expansive along v_i . The threshold is the separatrix at which posterior variance exceeds the injected noise variance:

$$a_i = \frac{\text{posterior variance along } v_i}{\text{noise variance } t} \quad \begin{cases} < 1 : & \text{denoising, contraction,} \\ = 1 : & \text{neutral,} \\ > 1 : & \text{amplification, expansion.} \end{cases} \quad (7)$$

The three design choices are then: threshold at 1 (only expansion is penalized), measure the *excess* $a_i - 1$ (the neutral point is 1, not 0), and square it (making the functional comparable to a squared Frobenius norm and penalizing strong expansion superlinearly).

Link to ambiguity. Suppose the posterior at y is approximately $\frac{1}{2}\mathcal{N}(\mu_1, \Sigma) + \frac{1}{2}\mathcal{N}(\mu_2, \Sigma)$, corresponding to two competing clean configurations. Then

$$\text{Cov}(X \mid Y_t = y) = \Sigma + \frac{1}{4}(\mu_1 - \mu_2)(\mu_1 - \mu_2)^\top, \quad (8)$$

and the between-mode term can exceed tI along $\mu_1 - \mu_2$, giving $a_i > 1$. So $a_i > 1$ signals that the noisy observation fails to identify a single locally stable latent configuration.

Remark 6.8 (Scope of the interpretation). Strictly, $a_i > 1$ is a diagnostic that posterior variance exceeds the noise scale. Multimodality is the typical, but not the only, mechanism: heavy-tailed unimodal p_0 (e.g. Laplace, Student- t) also produces $a_i > 1$. “Ambiguity” is the geometric reading of the phenomenon, not a theorem.

Remark 6.9 (Mean versus tail). $\mathcal{A}$ is an average over $y \sim p_t$ and can be small while $\sup_y a_{\text{max}}(y)$ is large, if the expansive region carries little mass. In practice one should report both $\mathcal{A}(t)$ and a high quantile of $(a_{\text{max}} - 1)_+$, since sampler failure is driven by trajectories that enter the expansive region, not by its measure.

6.2 Experiments

References

Peter Potaptchik, Iskander Azangulov, and George Deligiannidis. Linear convergence of diffusion models under the manifold hypothesis. *arXiv preprint arXiv:2410.09046*, 2024.